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FINAL YEAR PROJECT PRESENTATION

CareShare

Medical Equipment Sharing Bank

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Department of Information Technology | Academic Year 2025–2026

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Introduction to CareShare

What is CareShare?

CareShare is an online Medical Equipment Sharing Bank designed to connect donors and borrowers for temporary medical equipment needs. The platform enables users to donate, request, and borrow equipment safely and efficiently.

Objective

- Reduce medical expenses
- Promote equipment reuse
- Help patients in emergencies
- Build a community-driven healthcare support system

Supported Medical Equipment

- Wheelchairs
- Walkers
- Crutches
- Patient Beds
- Oxygen Stands
- Commode Chairs
- Bedside Tables
- Walking Frames

Problem Statement

Many patients require medical equipment only for a short period after surgery or during recovery — yet no trusted system exists to share it.

Challenges

- Medical equipment is expensive
- Many families cannot afford temporary purchases
- Used equipment often remains unused after recovery
- Lack of a trusted platform for sharing equipment
- Difficulty in finding nearby available equipment
- No proper verification or tracking system

Result



Financial Burden



Resource Wastage



Limited Accessibility



Increased Medical Costs

Proposed Solution

CareShare provides a centralized platform where people can donate or borrow medical equipment securely.

- **Equipment Listing**
- **Borrow Equipment**
- **Donor Registration**
- **Borrower Requests**
- **Equipment Verification**
- **Sanitization Status**
- **Return Tracking**
- **Availability Tracking**
- **User Authentication**
- **Community Support**

Outcome

- **Affordable healthcare support**
- **Reduced equipment wastage**
- **Better accessibility**
- **Strong community collaboration**

System Overview

Frontend

Responsible for user interface and interaction.

- User Dashboard
- Equipment Listing
- Request Forms
- Search & Filter
- Responsive Design

Backend

Responsible for application logic.

- REST APIs
- Equipment Management
- Request Processing
- Return Tracking
- Authentication

Database

Stores application data.

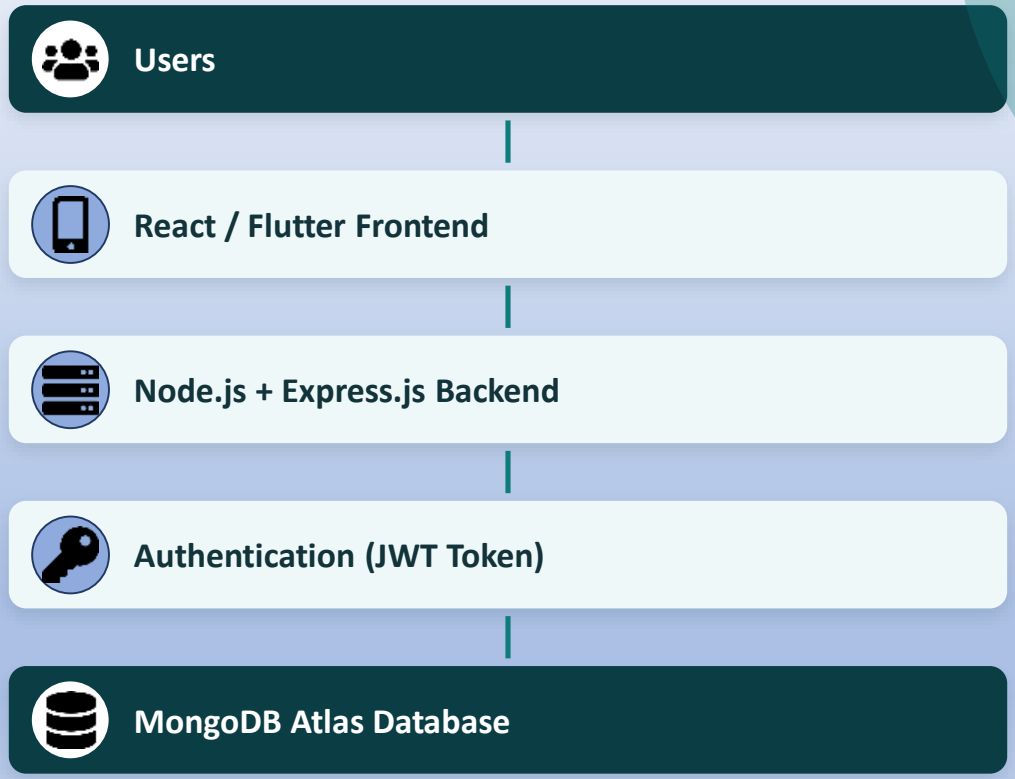
- User Information
- Equipment Details
- Borrow Requests
- Return Status
- Donation Records

Authentication

Secure login system.

- JWT Authentication
- User Registration
- Login
- Authorization
- Protected Routes

System Architecture



Technology Stack



Frontend

HTML5, CSS3, JavaScript, Bootstrap, React.js, TypeScript, Flutter (Mobile)



Backend

Node.js, Express.js



Database

MongoDB Atlas



Authentication

JWT — JSON Web Token



Version Control

Git & GitHub



API Testing

Postman

Key Features

Equipment Listing

Users can list available medical equipment.

Donor Registration

Anyone can donate unused equipment.

Borrow Equipment

Patients can request equipment.

Search & Filter

Find equipment quickly.

Condition Verification

Verify equipment quality before sharing.

Sanitization Status

Display cleaned and sanitized equipment.

Return Tracking

Track borrowed equipment until returned.

Secure Login

JWT-based authentication system.

Community Support

Encourages sharing and social responsibility.

Functional Requirements



User Module

- User Registration
- User Login
- User Profile



Equipment Module

- Add Equipment
- Edit Equipment
- Delete Equipment
- Availability



Request Module

- Send Borrow Request
- Accept / Reject
- Return Request



Admin Module

- Manage Users
- Verify Equipment
- Monitor Requests
- Generate Reports



Security Module

- JWT Authentication
- Protected APIs
- User Authorization

Benefits of CareShare

- Affordable medical support
- Reuse of unused equipment
- Reduces medical waste
- Supports needy patients
- Faster equipment availability
- Secure online platform
- Easy donor & borrower communication
- Community welfare
- Environment-friendly solution

Future Enhancements

- Mobile Application
- GPS-based Nearby Search
- QR Code Equipment Tracking
- Online Chat Support
- AI Equipment Recommendation
- Payment Gateway for Deposit
- Hospital Integration
- NGO Integration
- Email & SMS Notifications
- Analytics Dashboard

Conclusion

CareShare is a smart, secure, and community-driven Medical Equipment Sharing Bank that helps patients access temporary medical equipment easily — contributing toward a more accessible, sustainable, and socially responsible healthcare ecosystem.

The system promotes

- **Affordable Healthcare**
- **Resource Optimization**
- **Equipment Reuse**
- **Community Support**
- **Digital Healthcare**



Thank You

Questions & Answers

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